INTRODUCTION · EXAM STRATEGY - ALL SECTIONS

How This Course Works

What the Professional Data Engineer exam looks like today, how this course was built, the company you will design for, and a route through eleven hours of material.

EXAM Professional Data Engineer MODULES 45 RUNTIME Over 11 hours

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The exam rarely asks what a product is. It asks which product fits, and why the others do not.

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The exam, as Google describes it today

- **2 hours**, **40 to 50** multiple-choice and multiple-select questions.
- English or Japanese. **Online**, or at a **test centre**.
- Valid for **2 years**. No prerequisites. Three years' experience recommended.
- The **pass mark is not published**. Any figure you read online is a guess.

> CURRENCY CHECK
>
> Checked against Google's certification page on 14 October 2026.

Five sections, and how the course spends its time

SECTION	GUIDE WEIGHT	TIME IN THIS COURSE
1 Designing data processing systems	22%	2 h 05, modules 02 to 09
2 Ingesting and processing the data	25%	2 h 09, modules 10 to 18
3 Storing the data	20%	1 h 52, modules 19 to 26
4 Preparing and using data for analysis	15%	1 h 20, modules 27 to 32
5 Maintaining and automating workloads	18%	1 h 35, modules 33 to 39

THE NUMBERS

Teaching time tracks the official weights to within about one percentage point per section.

Renewing in two years

- Sit the **standard exam** again, as for the first time.
- Or a **renewal exam**: 1 hour, 20 questions, during the renewal window.
- Or **Google Skills** courses and skill badges, in the last year of your certificate.
- A renewal exam lasts **2 years**. A Google Skills renewal lasts **1 year**.

> CURRENCY CHECK
>
> Google Skills is the new name for Google Cloud Skills Boost.

Old names on the exam, new names in the console

- Google's banner: the exam "will soon be updated to reflect recent branding changes".
- The guide still says **Dataproc**, **Cloud Composer**, **BigLake** and **Dataplex**.
- The console now says Managed Spark, Managed Airflow, Lakehouse and Knowledge Catalog.
- This course teaches **both names**, every time. Module 40 lists them all.

> THE PATTERN
>
> Answer with the name the question uses. A renamed product is still right. A retired one is not.

How this course was made

- A bank of practice questions showed **which topics and decisions** the exam tests.
- **None** of those questions is reused. Every scenario and paper is new.
- Every fact was checked against **Google Cloud documentation**, in October 2026.
- Where a page could not be checked, the course says so: **Honest limit**.

> REMEMBER
>
> Topics the bank missed, such as embeddings and Managed Kafka, were added from the current guide.

Meet the company you will design for

- **Larkspur Energy**: a fictional supplier in Bristol, with a German arm in Hamburg.
- **3.1 million** smart meters, sending a reading every half hour.
- Moving off **Oracle billing**, a **40-node Hadoop** cluster, and spreadsheets.
- German customer data must stay **in Germany**. UK work runs in **London**.

> FOR LARKSPUR
>
> Partners: Westgrid, the grid operator, wants demand within five minutes. Hearthly wants a private match.

How every module runs

01
The core idea
One sentence to hold on to

02
Teaching slides
Compares, tables, real code

03
Phrases
Words in a question that pick the answer

04
Worked scenarios
Pause, decide, then hear why

05
In five lines
The module, compressed

REMEMBER

When a scenario says pause now, do pause. Choosing is where the learning happens.

What the callouts mean

CALLOUT	WHAT IT TELLS YOU
The signal	The phrase in a question that decides the answer
The trap	A tempting wrong answer, and why it is wrong
The tiebreaker	How to choose between two options that both work
Currency check	Something renamed, retired or new since older courses
Honest limit	Something the docs did not confirm, or a figure to treat with care

THE PATTERN

Traps and tiebreakers are where most marks are won and lost.

The map, and Section 1: designing systems

MODULE	STARTS AT
01 The Google Cloud data map	0:00:00
02 IAM and organisation policies, 03 Encryption and keys	0:17:19
04 Personal data, 05 Residency and perimeters	0:51:33
06 Reliability and fidelity, 07 Migrating files	1:21:22
08 Migrating databases, 09 Portability and multi-cloud	1:52:49

REMEMBER

Start with module 01. It names every product in the course on one page.

Section 2: ingesting and processing

MODULE	STARTS AT
10 Planning pipelines, 11 Pub/Sub	2:22:35
12 Dataflow and Beam, 13 Windows	2:54:38
14 Watermarks and triggers, 15 Spark and Hadoop	3:21:03
16 Getting data into BigQuery, 17 Low-code and SQL pipelines	3:49:04
18 Deploying pipelines	4:18:46

THE TIEBREAKER

Modules 12 to 14 are the hardest stretch. Watch them in order, and rewatch 14 if late data feels slippery.

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Section 3: storing the data

MODULE	STARTS AT
19 Choosing a storage service, 20 Bigtable	4:31:54
21 Relational stores, 22 Cloud Storage	5:01:26
23 BigQuery tables, 24 Modelling the warehouse	5:28:47
25 Lakes and lakehouses, 26 Knowledge Catalog	5:57:14

REMEMBER

Module 19 is the one to rewatch the night before. It decides between every store.

Sections 4 and 5: analysis, and operations

MODULE	STARTS AT
27 Serving BI, 28 Slow queries, 29 Securing BigQuery	6:23:49
30 BigQuery ML, 31 Embeddings and RAG, 32 Sharing	7:06:30
33 Capacity, 34 Right-sizing, 35 Orchestration	7:43:28
36 Monitoring, 37 Operating streaming	8:22:47
38 When pipelines fail, 39 Replication and failover	8:49:05

CURRENCY CHECK

Module 31 covers the newest material in the guide: embeddings, vector search and RAG.

Exam craft and the practice papers

MODULE	STARTS AT
40 What changed: names, retirements, guide vs today	9:18:08
41 Confusion pairs: the one word that separates them	9:29:27
42 How PDE questions are written	9:49:06
43 Practice paper 1, 20 questions	10:01:27
44 Practice paper 2, 45 Practice paper 3	10:27:48

THE NUMBERS

Sixty new questions in all. Paper 3 ends with a way to read your three scores together.

A route through the course

01

Modules 01 to 39

In order, pausing on every scenario

02

Modules 40 to 42

Names, pairs, question craft

03

Paper 1

Score it by section

04

Fix the gap

Rewatch the weakest section

05

Papers 2 and 3

Then read all three scores

THE TIEBREAKER

Short on time? Never skip the scenarios. They are the closest thing to the real exam.

Introduction in five lines

- **2 hours**, 40 to 50 questions, five weighted sections, no published pass mark.
- The exam uses **old names**. This course teaches both, every time.
- Built from what the exam tests, with **no reused questions** and checked docs.
- You design for **Larkspur Energy**, its meters, and its German data.
- Pause on every scenario. **Choosing** is where the learning happens.

> COMING UP
>
> Next: module 01, the Google Cloud data map, and the 2026 names.